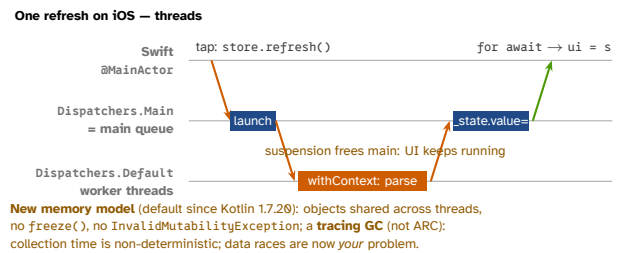
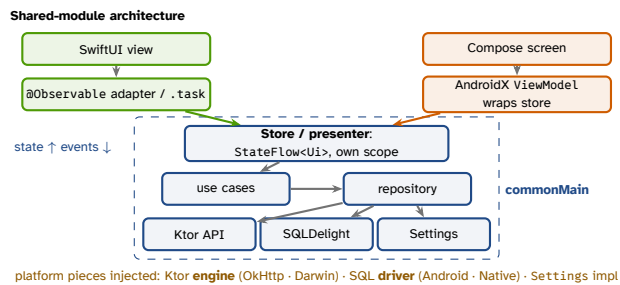


In one line: A shared module is **repositories + use cases + a state holder exposing StateFlow** in commonMain, built on **Ktor**, **kotlinx.serialization**, **SQLDelight**, **Koin** and **kotlinx-datetime**, with per-platform engines and drivers; each platform binds native UI to it. On iOS, Dispatchers.Main is the main queue and the **new memory model** (tracing GC, no freezing) lets objects cross threads like on the JVM.



The stack — common core, platform piece

Library	Role · platform part	iOS twin
Ktor client	HTTP, plugins (ContentNegotiation, auth, logging); engine per target : OkHttp/Android · Darwin (NSURLSession)	URLSession
kotlinx .serialization	@Serializable + compiler plugin , no reflection; Json { ignoreUnknownKeys = true }; sealed = class discriminator	Codable
SQLDelight	write .sq SQL → typed Kotlin; .sqm migrations verified; asFlow(); driver: AndroidSqliteDriver · NativeSqliteDriver	GRDB / Core Data
Koin · kotlin-inject	runtime DSL (errors at runtime) · KSP compile-time graph; Hilt is Android-only	init injection
kotlinx -datetime	Instant, LocalDateTime, TimeZone — no java.time in common	Date, Calendar
multiplatform -settings	KV over SharedPreferences / UserDefaults — not for secrets (Keychain via an interface)	UserDefaults
Kermit / Napier	logging to Logcat / os_Log	Logger

Exposing state to SwiftUI

- Shared store owns CoroutineScope(SupervisorJob() + Dispatchers.Main) and a clear(); Android calls it from onCleared, iOS from the owner's teardown — iOS has no viewModelScope.
- Swift side: **SKIE** (for await over the flow), **KMP-NativeCoroutines** (asyncSequence(for:)), or a hand-written watch({...}) → cancel handle wrapper. Map Kotlin types to Swift structs in the adapter.

```

class DevicesStore(private val repo: DeviceRepo,
    private val scope: CoroutineScope) { // injected
    private val _s = MutableStateFlow<Ui>(Ui.Loading)
    val state: StateFlow<Ui> = _s.asStateFlow()
    fun refresh() { scope.launch {
        _s.value = try { Ui.Loaded(repo.devices()) }
        catch (e: ApiError) { Ui.Failed(e.message) } } }
    fun clear() = scope.cancel()
}
  
```

```

.task { for await s in store.state { ui = map(s) } } // SKIE
  
```

Threading & memory on iOS

- Dispatchers.Main on iOS = main dispatch queue (in coroutines-core; Android needs -android). Dispatchers.IO exists on Native since coroutines 1.7 — still inject dispatchers.
- Old model (pre-1.7.20): shared objects had to be **frozen** (immutable) or crashed — the #1 historic KMP pain; now gone.
- Hop to @MainActor before touching SwiftUI state; never runBlocking on the main thread.
- Kotlin objects in Swift are GC-managed: deinit-style cleanup is late — call close()/clear() explicitly; capture [weak self] in closures handed to Kotlin.

Testing shared code

- commonTest + kotlin.test; runTest = virtual time (skips delay); StandardTestDispatcher + Dispatchers.setMain; **Turbine** flow.test { awaitItem() } for hot flows.
- Fakes over mocks (MockK is JVM-only); Ktor MockEngine; in-memory SQLite driver. JVM runs fast on CI; iOS-simulator tests need macOS.

Team & process

- Ownership:** a cross-platform core team, or iOS becomes a second-class consumer of an Android-shaped API. iOS devs **review the Swift surface**.
- Debugging:** iOS devs must read Kotlin; Xcode steps into Kotlin only with a plugin (Touchlab xcode-kotlin); Kotlin/Native link time slows iOS builds.
- Delivery:** monorepo + build phase (fast iteration) vs versioned XCFramework via SPM (decoupled, slower loop). Crash reports need Kotlin symbolication.
- Adopt incrementally:** one module (API client, validation) behind a facade; measure, then grow.

Traps

- ! Forgot the iOS Ktor engine / SQL driver → compiles, fails at runtime on iOS.
- ! catch (e: Exception) around suspend eats cancellation — rethrow.
- ! StateFlow is conflated: rapid events are lost — a Channel for events.
- ! Hard-coded Dispatchers.Main in shared code = untestable on the JVM.

Remember

Core in common, engines and drivers per platform · inject scope and dispatcher · no freezing, but a GC · iOS reviews the API.